

Turbulence Analysis Report -

2026-02-22

Generated: 2026-02-22 17:19:26

Table of Contents

Interpretation of Cross-correlations / Anisotropy Index

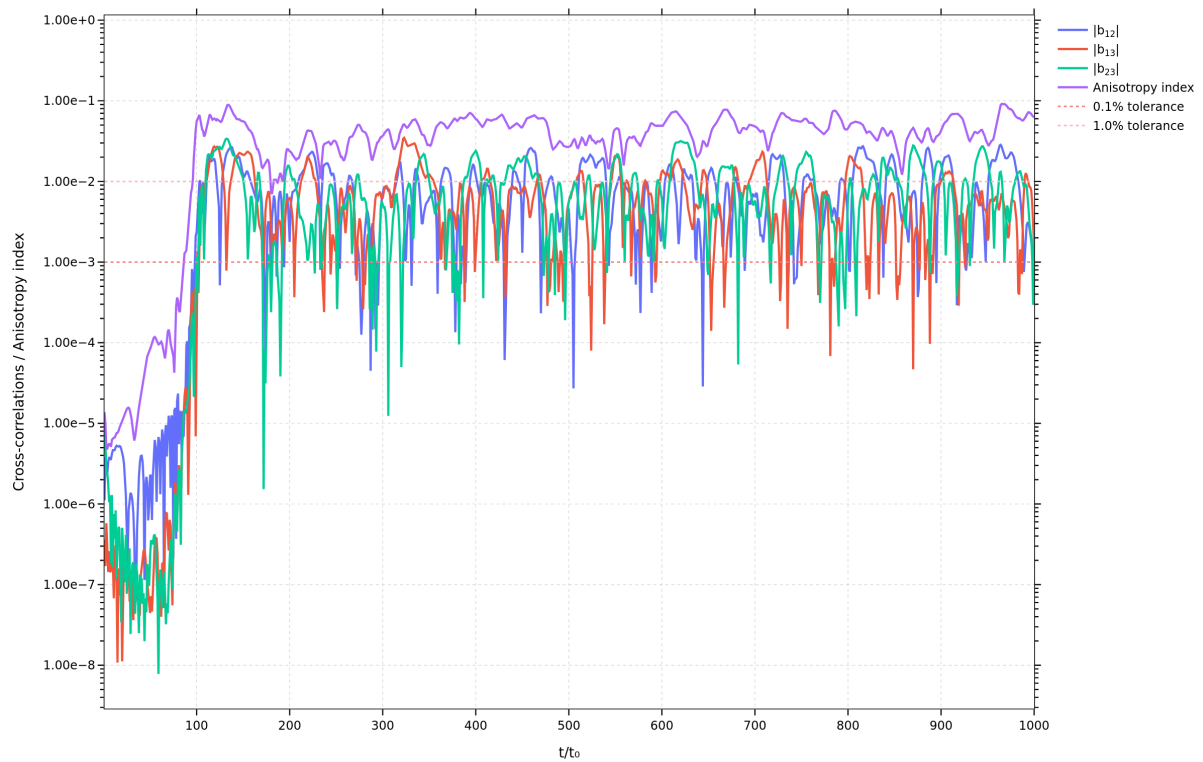


Figure 1. Evolution of cross-correlations and anisotropy index over normalized time.

Interpretation of Cross-correlations / Anisotropy Index

This figure, "Cross-correlations / Anisotropy index vs t/t_0 ", displays the evolution of off-diagonal components of the anisotropy tensor and a global anisotropy index over time, normalized by the initial time (t/t_0). The y-axis is on a logarithmic scale, highlighting the magnitudes of these quantities.

Here's a breakdown of the traces: * **$|b_{12}|$ (blue line), $|b_{13}|$ (red line), $|b_{23}|$ (green line):** These represent the absolute values of the off-diagonal components of the Reynolds stress anisotropy tensor. These components quantify the correlation between different velocity fluctuations (e.g., $u'v'$, $u'w'$, $v'w'$). In a perfectly isotropic turbulent flow, these values would be zero. * **Anisotropy index (purple line):** This is a scalar measure of the overall anisotropy of the turbulence, typically derived from the second and third invariants of the anisotropy tensor. A higher value indicates a greater departure from isotropy. * **0.1% tolerance (dashed red line) and 1.0% tolerance (dashed pink line):** These horizontal lines serve as reference thresholds, indicating levels of anisotropy or cross-correlation.

Interpretation:

- 1. Initial Phase ($t/t_0 < \sim 100$):** All cross-correlation components and the anisotropy index start at very low values (below $1.0e-5$), suggesting an initial state that is either highly isotropic or still developing.
- 2. Development Phase ($t/t_0 \sim 100-200$):** There is a sharp increase in all quantities. The anisotropy index rises significantly, reaching values around $1.0e-1$, while the cross-correlation components also increase, fluctuating between $1.0e-3$ and $1.0e-2$. This indicates the development of anisotropic features in the turbulent flow.
- 3. Quasi-Steady State ($t/t_0 > \sim 200$):**
 - The **Anisotropy index** (purple line) generally fluctuates between approximately $5.0e-2$ and $1.0e-1$, consistently staying above the 1.0% tolerance level. This suggests that the flow maintains a notable degree of anisotropy throughout the simulation.

- The **cross-correlation components** ($|\mathbf{b}_{12}|$, $|\mathbf{b}_{13}|$, $|\mathbf{b}_{23}|$) show significant fluctuations, often dropping to very low values (below $1.0\text{e-}4$) and peaking around $1.0\text{e-}2$. While they frequently dip below the 0.1% tolerance ($1.0\text{e-}3$), they also periodically exceed it, sometimes approaching the 1.0% tolerance. This indicates that while the individual correlations between specific velocity components can be small at times, they are not consistently negligible. The fluctuations suggest a dynamic and evolving anisotropic state rather than a perfectly steady one.

In summary, the figure illustrates that the simulated turbulence quickly develops anisotropic characteristics, with the overall anisotropy index remaining relatively high. While the individual cross-correlations between velocity components fluctuate significantly, they generally remain below the overall anisotropy index, indicating that the flow is not perfectly isotropic but also not dominated by strong, persistent correlations between any single pair of off-diagonal components.